

**Fourth Semester B. Tech. (Electronics and Communication
Engineering / Electronics Design Technology) Examination**

INTRODUCTION TO ELECTROMAGNETIC THEORY

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Assume suitable data wherever necessary. Consider bold letters as a vector.
- (3) Given Constants : $\epsilon_0 = 8.854 \times 10^{-12}$ F/m and $\mu_0 = 4\pi \times 10^{-7}$ H/m.

1. Solve any **Six** questions :

- (A) Given points A(2, 5, -1), B(3, -2, 4) and C(-2, 3, 1).
Find :
 - (i) The length of the projection of $\mathbf{R}_{AB}$ on $\mathbf{R}_{AC}$.
 - (ii) The angle between vector $\mathbf{R}_{AB}$ and $\mathbf{R}_{AC}$. 2(CO1)
- (B) Transform vector $5\mathbf{a}_x$ into cylindrical coordinate system at point P($\rho = 4$, $\phi = 120^\circ$, $z = 2$). 2(CO1)
- (C) Find the total charge contained in a spherical shell, if this thick spherical shell centered at the origin extends between $R = 2$ cm to $R = 3$ cm and volume charge density $\rho_v = 3R \times 10^{-4}$ C/m³. 2(CO2)
- (D) Elaborate Faraday's experiment. How the idea of flux density originate from it ? 2(CO2)
- (E) Elaborate the Biot-Savart law and its distributed form. 2(CO3)
- (F) Express Laplace operator in spherical coordinate system. 2(CO1)
- (G) Explain magnetization and give importance of magnetic Permeability. 2(CO3)
- (H) How polarization is significant in wave propagation ? 2(CO4)

2. (A) Planes $x = 2$ and $y = -3$ respectively, carry charges 10 nC/m^2 and 15 nC/m^2 . If the line charge $x = 1, z = 1$, carries charge $10\pi \text{ nC/m}$. Calculate $\mathbf{E}$ at $(2, 2, -2)$ due to the three charge distributions. 4(CO2)
- (B) Give significance of electric field intensity in distributed form and write the expression for it. 3(CO2)
- (C) Given the electric flux density $\mathbf{D} = 0.3 r^2 \mathbf{a}_r \text{ nC/m}^2$ in free space where $\epsilon_r = 3$. Find $\mathbf{E}$ at point $P(r = 2, \theta = 25^\circ \text{ and } \phi = 90^\circ)$. 3(CO1,2)
3. (A) Given the field $\mathbf{D} = 6\rho \sin 0.5\phi \mathbf{a}_\rho + 1.5\rho \cos 0.5\phi \mathbf{a}_\phi \text{ C/m}^2$. Evaluate either side of the divergence theorem for the region bounded by $\rho = 2, \phi = 0, \phi = \pi, Z = 0 \text{ and } z = 5$. 4(CO1,2)
- (B) Point charge of $+3 \mu\text{C}$ and $-3 \mu\text{C}$ are located at $(0, 0, 1) \text{ mm}$ and $(0, 0, -1) \text{ mm}$ respectively in free space. Find :
 (i) Dipole moment $\mathbf{p}$,
 (ii) $\mathbf{E}$ in spherical coordinate at $P(2, 40^\circ, 50^\circ)$. 3(CO1,2)
- (C) State boundary conditions between perfect dielectric materials. 3(CO4)
4. (A) A current sheet $\mathbf{K} = 8\mathbf{a}_x \text{ A/m}$ flows in the region $-2 < y < 2 \text{ m}$ in the plane $z = 0$. Calculate $\mathbf{H}$ at $P(0, 0, 3)$. 3(CO2,4)
- (B) Evaluate either side of stokes theorem for the portion of cone $\theta = 0.1\pi$ bounded by $r = 2, r = 4, \phi = 0 \text{ to } \phi = 0.3\pi$. For $\mathbf{H} = 6 r \sin\phi \mathbf{a}_r + 18r \sin\theta \cos\phi \mathbf{a}_\phi$. 4(CO1,2,4)
- (C) State Ampere circuital law and give its significance. 3(CO1,4)
5. (A) Obtain vector Helmholtz equation from phasor equation for electric field. 4(CO4)
- (B) Assume $\mu_1 = 4 \mu\text{H/m}$ in region - 1, where $z > 0$, while $\mu_2 = 7 \mu\text{H/m}$ where $z < 0$. Let $\mathbf{K} = 80 \mathbf{a}_x \text{ A/m}$ on the surface $z = 0$. The magnetic field $\mathbf{B}_1$ in the region - 1 is $\mathbf{B}_1 = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z \text{ mT}$. Find out the value of $\mathbf{B}_2$. 3(CO3,4)

- (C) Ampere circuital law for static magnetic field has form $\text{Curl } \mathbf{H} = \mathbf{J}$. When it is extended to time varying field it changes its form, hence give the significance of displacement current density $\mathbf{J}_d$. 3(CO3)

6. Write short notes on any **Two** :—

- (i) Significance of terms in power theorem.
- (ii) Polarization : linear and circular.
- (iii) Significance of zero and infinite potential.
- (iv) Vector magnetic potential and significance. 8(CO1,2,3,4)

